- Notations:

(1) Integer partition: $\lambda = (\lambda_1, \dots, \lambda_\ell)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell \geq 0$

(2) $|\lambda| := \lambda_1 + \lambda_2 + \cdots + \lambda_\ell$, $\ell(\lambda) :=$ # positive parts in λ

(3) λ^* : transpose of λ

(4) (French style) Ferrers shape of $\lambda := \{(i,j) \mid 1 \leq j \leq \ell(\lambda),\ 1 \leq i \leq \lambda_j\}$

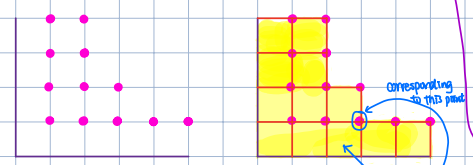
(5) lattice squares (boxes) of λ = boxes with NE corner in Ferrers shape of λ

(6) skew diagram : difference $\nu = \lambda/\mu$ of partition diagrams $\mu \subseteq \lambda$ or any translate of such a diagram (this allows the x-coordinates α_i and β_i of the left and right endpoints of some rows to be negative).

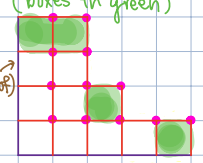
(7) content of a box $a = (i,j)$ = $i - j$.
ith row from left; jth column from bottom

Example: $\lambda = (5,3,2,2,0,0,0)$. We have • $|\lambda| = 12$, $\ell(\lambda) = 4$

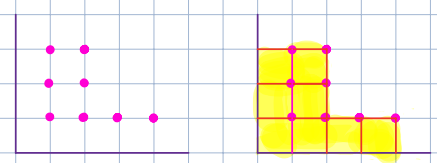
• Ferrers shape: • Boxes:



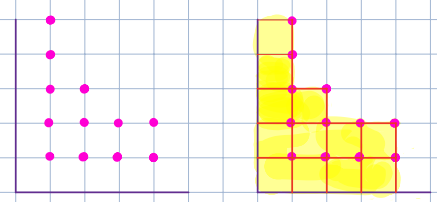
$\therefore \lambda/\mu$: (boxes in green)



$\mu = (4,2,2,0,0,0,0)$ • Ferrers shape: • Boxes:



$\lambda^* = (4,4,2,1,1)$ • Ferrers shape: • Boxes:



(8) $k = \mathbb{Q}(q,t)$

(9) $\Lambda = \Lambda_k(X)$ ($X = x_1, x_2, \dots$ infinite variables) : algebra of symmetric functions

(10) $\omega : \Lambda \to \Lambda$ s.t. $s_\lambda \mapsto s_{\lambda^*}$ (λ^*: conjugate of λ)

(11) $\langle\ ,\ \rangle : \Lambda \times \Lambda \to k$ s.t. $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$

(12) $n(\mu) = \sum_i (i-1)\mu_i$

(13) Modified Macdonald polynomial : $\tilde{H}_\mu(X; q, t) = t^{n(\mu)} \overbrace{J_\mu\left[\frac{X}{1-t^{-1}}; q, t^{-1}\right]}^{\text{Jack(q,t): Macdonald polynomial}}$

(14) ∇ : linear operator s.t. $\nabla \tilde{H}_\mu = t^{n(\mu)} q^{n(\mu^*)} \tilde{H}_\mu$

(15) $M = (1-q)(1-t)$

Example: • $n(5,3,2,2) = 0 \times 5 + 1 \times 3 + 2 \times 2 + 3 \times 2 = 13$

• $\nabla \tilde{H}_{5322} = t^{n(5322)} q^{n(4422110)} \tilde{H}_{5322} = t^{13} q^{15} \tilde{H}_{5322}$

- Plethystic evaluation

• A : any expression in terms of indeterminates (e.g. polynomial, rational function, formal series) — the variables q, t in our ground field also count as indeterminates.

$p_k[A]$ means substituting any indeterminate 'a' in A by 'a^k'.

e.g. $p_k[x_1+x_2+\cdots+x_\ell] = x_1^k+x_2^k+\cdots+x_\ell^k = p_k(x_1,\dots,x_\ell) \Rightarrow f[x_1+\cdots+x_\ell] = f(x_1,\dots,x_\ell) = f(X)\ \forall f\in\Lambda$.

Hence we may write $f[X]$ as another way to express $f(X)$, but in the bracket, $X = x_1+\cdots+x_\ell$.

e.g. $\because\ \omega p_k = (-1)^{k-1}p_k \Rightarrow \omega f(X) = f[-zX]\big|_{z=-1}$ $\left(p_k[-zX] = -z^k x_1^k - z^k x_2^k - \cdots = -z^k p_k(X) \Rightarrow p_k[-zX]\big|_{z=-1} = (-1)^{k-1}p_k(X) = \omega p_k(X) \right)$.

$$\sum_{k>0}\frac{x^k}{k} = -\ln(1-x) \Rightarrow \exp\sum_{k>0}\frac{x^k}{k} = \frac{1}{1-x} \Rightarrow \exp\sum_{k>0}\left(\frac{a_1^k+a_2^k+\cdots}{k} - \frac{b_1^k+b_2^k+\cdots}{k}\right) = \frac{\exp\sum_{k>0}\frac{a_1^k}{k}\cdot\exp\sum_{k>0}\frac{a_2^k}{k}\cdots}{\exp\sum_{k>0}\frac{b_1^k}{k}\exp\sum_{k>0}\frac{b_2^k}{k}\cdots} = \frac{\frac{1}{1-a_1}\cdot\frac{1}{1-a_2}\cdots}{\frac{1}{1-b_1}\cdot\frac{1}{1-b_2}\cdots} = \frac{\prod_i(1-b_i)}{\prod_i(1-a_i)}.$$

Let $\Omega = 1+\sum_{k>0} h_k = \exp\sum_{k>0}\frac{p_k}{k}$. We have $\Omega[a_1+a_2+\cdots-b_1-b_2-\cdots] = \frac{\prod_i(1-b_i)}{\prod_i(1-a_i)}$

(Classical) Cauchy identity: $h_k(XY) = \sum_{\lambda\vdash k} s_\lambda(X)s_\lambda(Y)$, or $\Omega[XY] = \sum_\lambda s_\lambda[X]s_\lambda[Y]$.

* By convention, $X = x_1, x_2, \dots$ stands for $x_1+x_2+\cdots$ inside a plethystic evaluation.

Then $f[X] = f[x_1+x_2+\cdots] = f(x_1,x_2,\dots) = f(X)$. $\underbrace{-(1-q^{\lambda_i}X(1-t^{\lambda_i}))}$ p_k

$\therefore$ For $f = \sum_\lambda c_\lambda p_\lambda$, $f[-tX] = \sum_\lambda c_\lambda p_\lambda[-tX] = \sum_\lambda c_\lambda p_{\lambda_1}[-tX]\cdots p_{\lambda_\ell}[-tX] = \sum_\lambda (-1)^{\ell(\lambda)} c_\lambda \prod_{i=1}^{\ell(\lambda)}(1-q^{\lambda_i}X(1-t^{\lambda_i}))\, p_\lambda(X)$